

Sem 4 HW

Review

$$y = 2x - x^2 \quad \text{At } x=4, y = 8 - 2 = 6$$

$$y' = 2 - 2x$$

$$\text{When } x=4, y' = 2 - 2(4) = -6 \\ \therefore \text{normal grad} = \frac{1}{3}$$

At point (4, 6):

$$y - 6 = \frac{1}{3}(x - 4)$$

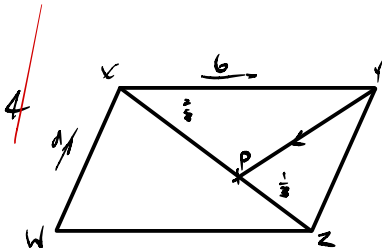
$$3y - 18 = x - 4$$

$$x - 3y + 14 = 0$$

2 d (stationary points)

$$3 y' = -2 \sin 2x^\circ$$

$$= -\frac{1}{180}$$



$$XZ = a$$

$$\frac{2}{3}XZ = \frac{2}{3}(a) = XP$$

$$XY = b$$

$$\therefore YP = YX + XP \\ = -b + \frac{2}{3}(a)$$

$$= -(\frac{2}{3}a + \frac{1}{3}b)$$

$$5 y 4x - x^2 \quad \text{when } y=0 \quad x=0 \text{ or } x=4$$

$$\therefore \int_0^4 (4x - x^2) dx$$

$$= \left[2x^2 - \frac{1}{3}x^3 \right]_0^4 = 32 - \frac{64}{3} \\ = \frac{32}{3} \text{ sq. units}$$

Consolidation

1 $a = 0.45$

$$r = \frac{1}{100}$$

$$S_{\infty} = \frac{0.45}{1 - \frac{1}{100}} \left(\frac{a}{1-r} \right)$$
$$= \frac{45}{99} = \frac{5}{11}$$

2 $a = 28$

$$r = \frac{1}{4}$$

$$S_{\infty} = \frac{a}{1-r}$$
$$= \frac{28}{\frac{3}{4}}$$
$$= \frac{112}{3}$$

3 i) 11 sessions

$$ii) S_{10} = \frac{11}{2} [2a + (n-1)d]$$
$$= \frac{11}{2} [20 + 20] = 220$$

4 i $18000 \times (1.04)^{10} = £26,644$

$$ii) S_{10} = \frac{18000(1-1.04^{10})}{1-1.04}$$
$$= £216110$$

5 a) has value V .

$$\text{Value after } n \text{ years} = V \times 0.8^n$$

$$V \times 0.8^n = \frac{1}{4} V$$

$$0.8^n = \frac{1}{4}$$

$$n \ln 0.8 = -\ln 4$$

$$n = -\frac{\ln 4}{\ln 0.8}$$

$$= 6.2113 \text{ years} \quad \therefore \text{keeps car for 7 years}$$

b) car could break down or someone could accidentally damage it. $\rightarrow$ mileage.

$$6 \ a = 200$$

$$r = 0.8$$

$$i) \ 200 \times 0.8^n = 10$$

$$0.8^n = \frac{1}{20}$$

$$n = -\frac{\ln 20}{\ln 0.8}$$

$$= 14 \text{ games}$$

$$ii) \text{ total } d = 200 + 2S_{\infty}$$

$$= 1800 \text{ cm}$$

$$= 18 \text{ m}$$

$$S_{\infty} = \frac{160}{1-0.8}$$

$$= 800$$

$$iii) \ S_n = \frac{160(1-r^n)}{1-r} + 200$$

$$= \frac{160(1-0.8^n)}{0.2} > 0.44 \times 800$$

$$n > 20.1$$

$$\therefore \text{ after 21 games } (n)$$

$$7 \ u_1 = 54$$

$$u_4 = 2 = 54r^3$$

$$r^3 = \frac{1}{27}$$

$$r = \frac{1}{3}$$

$$u_n = 54 \times \left(\frac{1}{3}\right)^{n-1}$$

$$S_{\infty} = \frac{54}{\frac{2}{3}} = 81$$

$$0.44 \times 81 > 80.14$$

$$S_n = \frac{54(1-\frac{1}{3}^n)}{1-\frac{1}{3}}$$

$$81(1-\frac{1}{3}^n) > 80.14$$

$$1-\frac{1}{3}^n > \frac{99}{100}$$

$$\frac{1}{3}^n > \frac{1}{100}$$

$$3^n > 100$$

$$n > \log_3 100$$

$$> 4.19$$

$$n = 5 \quad (n \in \mathbb{N})$$

Exam Questions

$$1 \quad r = \frac{3}{4} \quad \left(\frac{3}{4}\right)^N \leq 0.0002 \quad S_{\infty} = \frac{12}{1-\frac{3}{4}} = 48$$

$$\therefore U_n = 12 \times \left(\frac{3}{4}\right)^{n-1}$$

$$S_{\infty} = 48 \quad S_N = \frac{12(1-\frac{3}{4}^N)}{(1-\frac{3}{4})} = 48(1-\frac{3}{4}^N)$$

$$48 - 48(1-\frac{3}{4}^N) \leq 0.0096$$

$$1 - (1-\frac{3}{4}^N) \leq 0.0002$$

$$\frac{3}{4}^N \leq 0.0002$$

$$2 \text{ a) } S_{25} = \frac{25}{2} [2 \times 10 + (25-1) \times 2] \\ = 850_p = £8.50$$

$$6 \text{ i) } 500 \times 1.03^n > 600 \\ 1.03^n > 1.2$$

$$n \log 1.03 > \log 1.2$$

$$n > \frac{\log 1.2}{\log 1.03}$$

$$> 6.2$$

$$\therefore n = 7 \text{ years}$$

ii) Alice takes no money out of the account.